

Comprehensive Analysis of Field Oriented Control (FOC) for a 3-Phase Induction Motor

Project Report Submitted in partial fulfilment of the requirements
for the degree of Bachelor of Technology from Maulana Abul
Kalam Azad University of Technology, West Bengal

By

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Under the Guidance of

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Certificate of Recommendation

It is hereby recommended to consider the project report entitled "Comprehensive Analysis of Field Oriented Control (FOC) for a 3-Phase Induction Motor" submitted by Joydip Karmakar for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electrical Engineering from Maulana Abul Kalam Azad University of Technology (formerly known as West Bengal University of Technology).

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Certificate of Approval

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Board of Examiners

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Abstract

This report provides a comprehensive, component-level technical analysis of the Simulink model implementing an Indirect Field Oriented Control (IFOC) architecture for a 50 HP Squirrel-Cage Induction Motor. Intended as an advanced instructional document for electrical machines and drives, this report systematically deconstructs the theoretical and practical frameworks embedded within the simulation. It begins by establishing the plant's equivalent circuit parameters and dynamic modelling. The analysis explores critical coordinate transformations required to map stationary three-phase variables into a synchronously rotating $d - q$ reference frame. By examining the system's decoupling equations, the report demonstrates how IFOC achieves independent regulation of rotor flux and electromagnetic torque. Furthermore, the document evaluates the discrete-time closed-loop control strategy and dissects the Space Vector Pulse Width Modulation (SVPWM) synthesis used to drive the three-phase inverter.

Chapter 1

Introduction

1.1 Introduction to Field Oriented Control

Traditionally, the control of induction machines was dominated by scalar methods, such as constant Volts-per-Hertz (V/f) control. While adequate for basic steady-state operations, scalar control governs only the magnitude and frequency of the applied voltage and currents, fundamentally ignoring the spatial orientation of the electromagnetic fields.

Field Oriented Control (FOC), also known as Vector Control, represents a revolutionary paradigm shift in AC machine drives, overcoming the limitations of scalar control. The fundamental objective of FOC is to emulate the decoupled control dynamics of a separately excited DC motor.

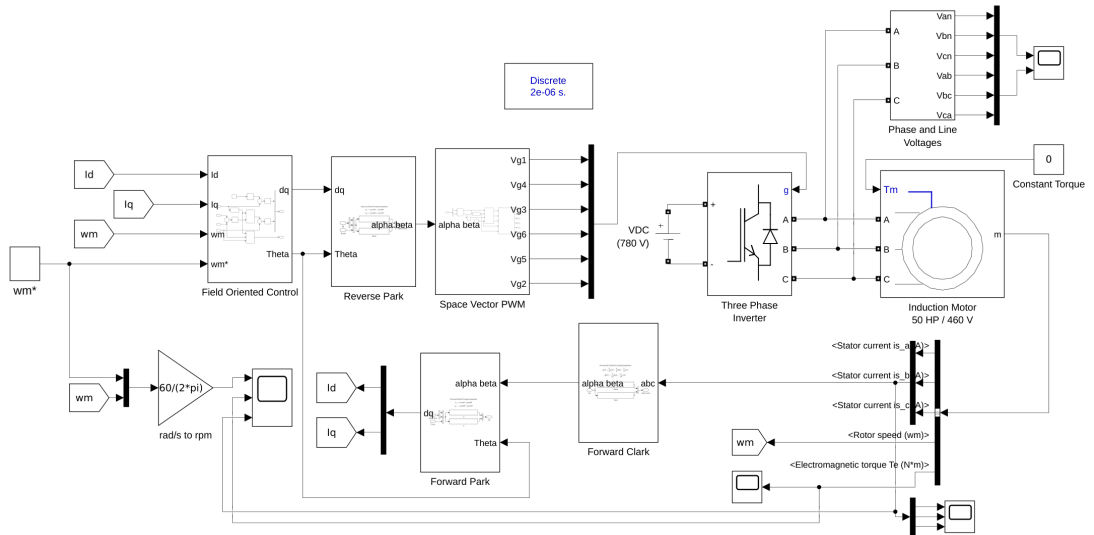


Figure 1.1: Top-Level Simulink Model Architecture of the FOC for Induction Motor

By deliberately aligning the direct-axis (d-axis) of this rotating frame with the rotor flux linkage vector, the overall stator current is mathematically decomposed into two entirely independent, orthogonal components:

- **Flux-Producing Component (i_{ds}):** The direct-axis stator current aligns with the rotor flux linkage vector (ψ_r).
- **Torque-Producing Component (i_{qs}):** The quadrature-axis stator current is maintained at a 90° electrical angle relative to the rotor flux axis.

The developed electromagnetic torque can be expressed via the decoupled equation:

$$T_e = \frac{3}{2} \left(\frac{P}{2} \right) \frac{L_m}{L_r} \psi_r i_{qs} \quad (1.1)$$

Substituting the motor parameters ($P = 4$, $L_m = 0.0347$ H, $L_r = 0.0355$ H), and assuming nominal flux $\psi_r = 0.96$ Wb:

$$T_e \approx 2.8152 \cdot i_{qs} \text{ N} \cdot \text{m} \quad (1.2)$$

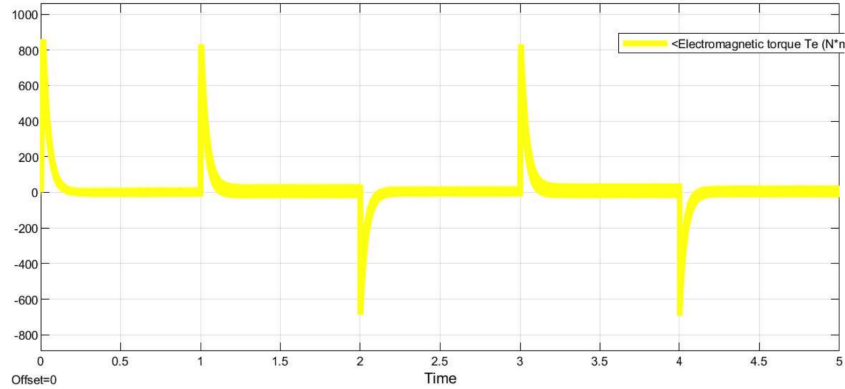


Figure 1.2: Transient and Steady-State Electromechanical Torque Tracking the Load Demands

Chapter 2

Induction Motor Plant

Parameters and Dynamic Modelling

2.1 Plant Parameters

The electromechanical plant governed by this control architecture is a three-phase Squirrel-Cage Induction Motor (SCIM). Based on the model's configurations, the motor is defined by the parameters summarised in Table 2.1.

Table 2.1: Squirrel-Cage Induction Motor Nominal Ratings and Equivalent Circuit Parameters

Symbol	Parameter Description	Value
<i>Nominal Ratings</i>		
S	Nominal Power	50 HP (37.3 kW)
V_s	Stator Line-to-Line Voltage	460 V
f	Nominal Frequency	50 Hz
P	Number of Poles	4 (2 Pole Pairs)
N_s	Synchronous Speed	1500 RPM
<i>Electrical Parameters</i>		
R_s	Stator Resistance	0.087 Ω
R_r	Rotor Resistance	0.228 Ω
L_{ls}	Stator Leakage Inductance	0.1 mH
L_{lr}	Rotor Leakage Inductance	0.8 mH
L_m	Magnetizing Inductance	34.7 mH
<i>Mechanical Parameters</i>		
J	Moment of Inertia	0.662 kg·m ²
F	Viscous Friction Coefficient	0.1 Nms/rad

The lumped self-inductances are derived as:

$$L_s = L_{ls} + L_m = 0.0348 \text{ H} \quad (2.1)$$

$$L_r = L_{lr} + L_m = 0.0355 \text{ H} \quad (2.2)$$

The rotor time constant (τ_r) is critical for Indirect FOC:

$$\tau_r = \frac{L_r}{R_r} = \frac{0.0355}{0.228} \approx 0.1557 \text{ s} \quad (2.3)$$

2.2 Mechanical Dynamics

The mechanical equation of motion dictating the rotor's dynamic trajectory is governed by:

$$T_e - T_L = J \frac{d\omega_m}{dt} + F\omega_m \quad (2.4)$$

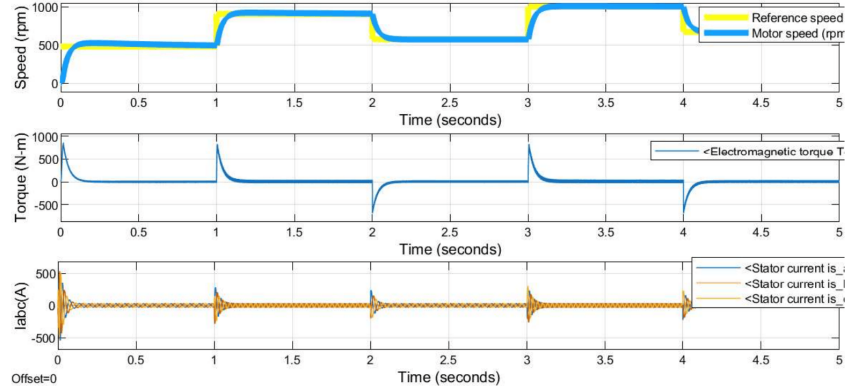


Figure 2.1: Overall Mechanical Dynamics Scope: Speed, Torque, and Stator Currents

Chapter 3

Control System Architecture

3.1 Coordinate Transformations

The central premise of FOC lies in transforming time-varying, three-phase AC stator variables into a synchronously rotating $d - q$ reference frame.

3.1.1 Forward Clarke Transform

Projecting measured stator phase currents onto a two-dimensional orthogonal reference frame ($\alpha - \beta$ axes):

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (3.1)$$

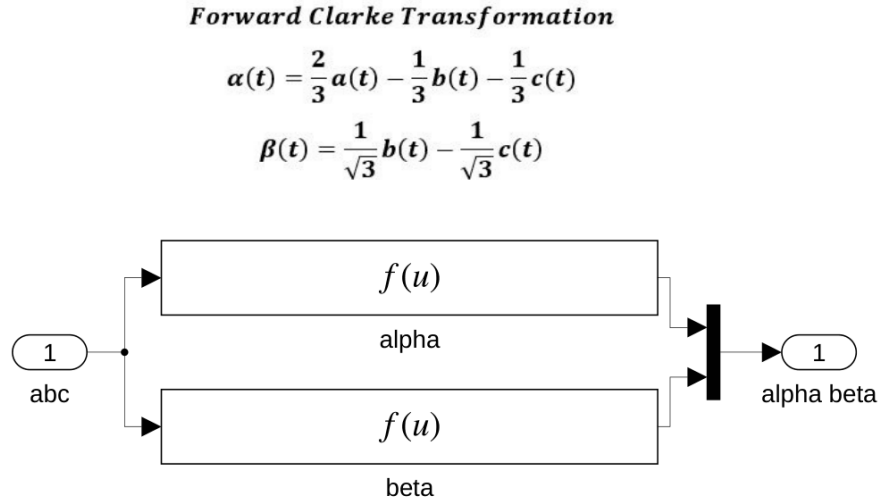


Figure 3.1: Forward Clarke Transform Subsystem

3.1.2 Forward Park Transform

Projecting the stationary $\alpha - \beta$ frame onto a synchronously rotating reference frame ($d - q$ axes):

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos \theta_e & \sin \theta_e \\ -\sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \quad (3.2)$$

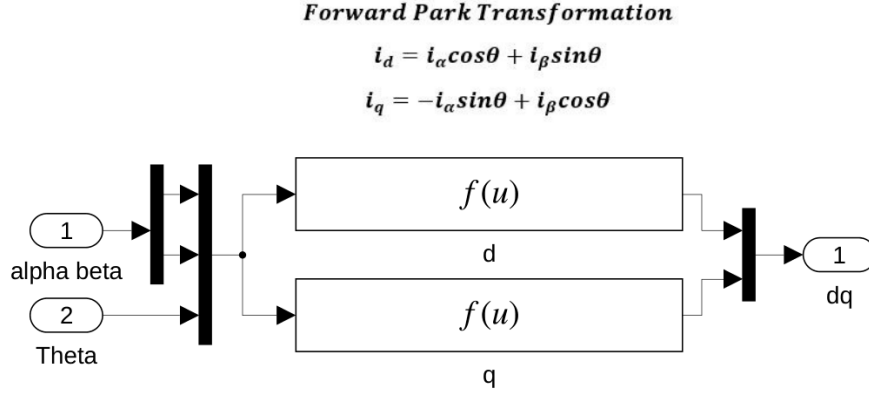


Figure 3.2: Forward Park Transform Subsystem

3.1.3 Reverse Park Transform

Translating required decoupling voltage commands (v_d^* and v_q^*) back to the stationary frame:

$$\begin{bmatrix} v_\alpha^* \\ v_\beta^* \end{bmatrix} = \begin{bmatrix} \cos \theta_e & -\sin \theta_e \\ \sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix} \quad (3.3)$$

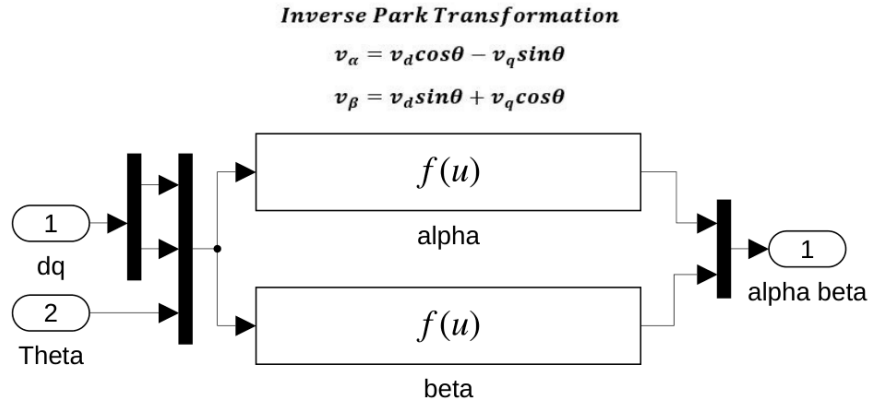


Figure 3.3: Reverse Park Transform Subsystem

3.2 Flux and Torque Decoupling Equations

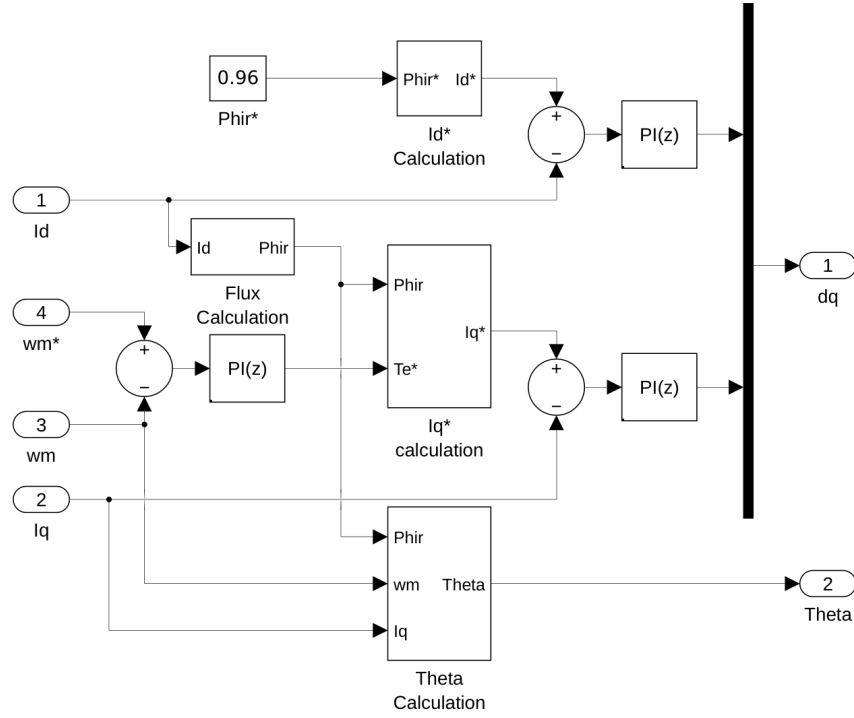


Figure 3.4: Detailed Implementation of the FOC Control Loops

The reference direct-axis current simplifies to a pure proportional relationship:

$$i_{ds}^* = \frac{\psi_r^*}{L_m} \quad (3.4)$$

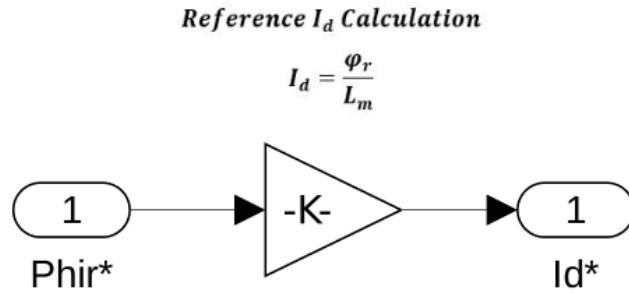


Figure 3.5: I_d^* Reference Calculation Block

The reference quadrature-axis current is calculated as:

$$i_{qs}^* = \frac{2}{3} \cdot \frac{2}{P} \cdot \frac{L_{lr} + L_m}{L_m} \cdot \frac{T_e^*}{\psi_r} \quad (3.5)$$

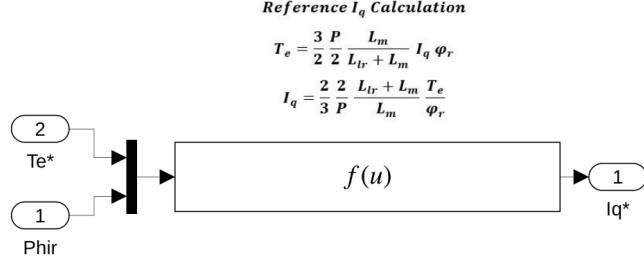


Figure 3.6: I_q^* Reference Calculation Block

To prevent a division-by-zero singularity during initial magnetisation, a numerical perturbation ($\epsilon = 1 \times 10^{-3}$) is added to the flux denominator.

3.3 Slip and Flux Angle Synthesis

In IFOC, the slip frequency (ω_{sl}) is dictated by:

$$\omega_{sl} = \frac{L_m}{\tau_r} \cdot \frac{i_{qs}^*}{\psi_r^*} \quad (3.6)$$

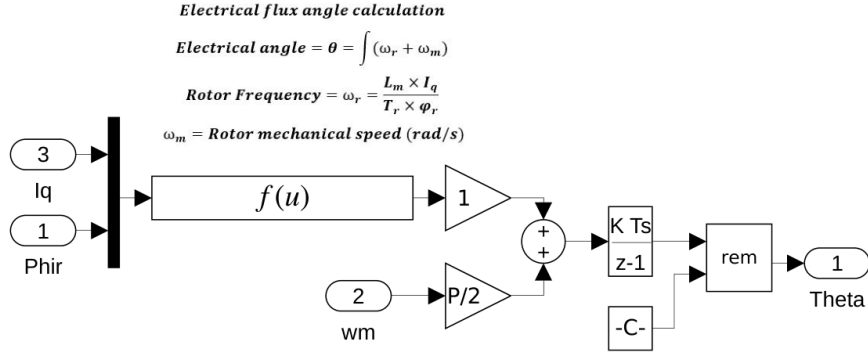


Figure 3.7: IFOC Slip and Flux Angle Estimation Subsystem

Chapter 4

Space Vector Pulse Width Modulation (SVPWM)

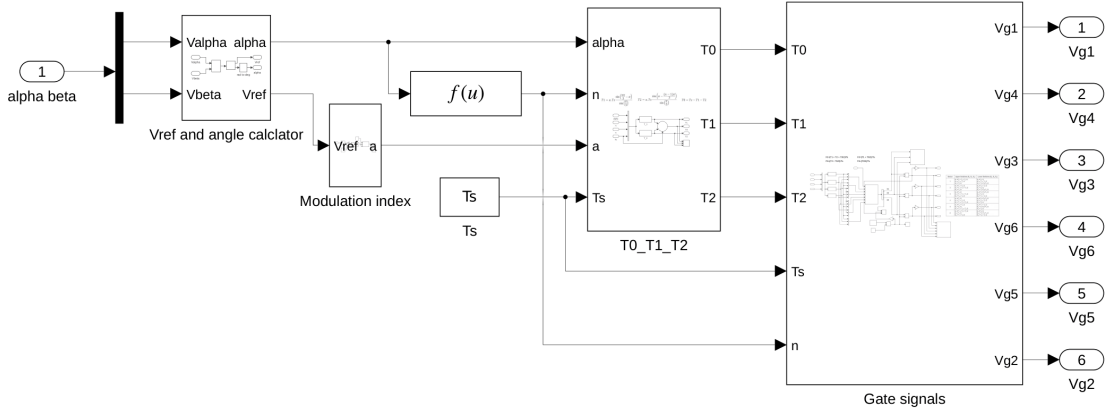


Figure 4.1: Main Architecture of the Space Vector PWM Generation Subsystem

SVPWM inherently injects a zero-sequence voltage into the phase commands to prevent premature clipping against the DC rails. This yields enhanced DC bus utilisation and reduced harmonic distortion.

4.1 Algorithm Execution

The incoming stationary frame voltage commands (v_α^* and v_β^*) define the reference vector $\vec{V}_{ref}$.

$$|\vec{V}_{ref}| = \sqrt{(v_\alpha^*)^2 + (v_\beta^*)^2}, \quad \theta_{ref} = \arctan\left(\frac{v_\beta^*}{v_\alpha^*}\right) \quad (4.1)$$

The dwell times for active vectors are calculated using the volt-second balance principle.

The modulation limits are enforced, and times are allocated to the inverter legs using symmetrical switching logic.

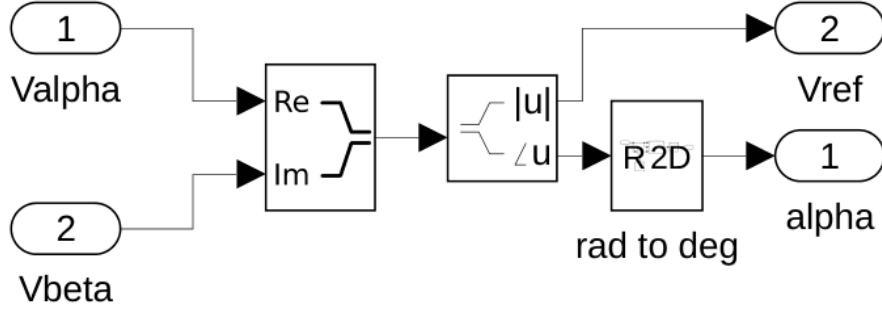


Figure 4.2: Transformation to Magnitude and Angle

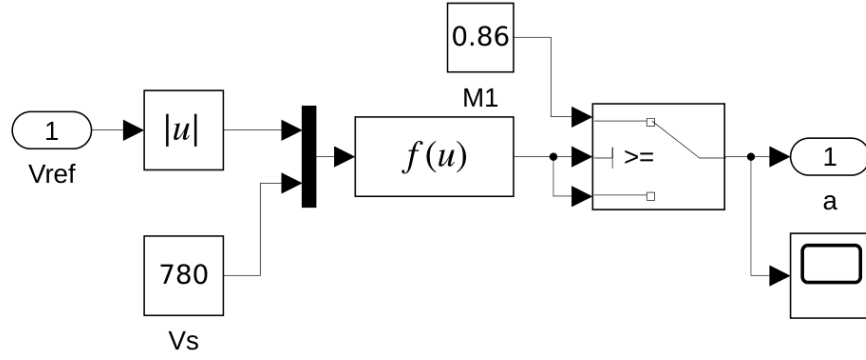


Figure 4.3: Modulation Index Calculation and Overmodulation Safeguard

$$T1 = a \cdot Ts \frac{\sin\left(\frac{n\pi}{3} - \alpha\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T2 = a \cdot Ts \frac{\sin\left(\alpha - \frac{(n-1)\pi}{3}\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T0 = Ts - T1 - T2$$

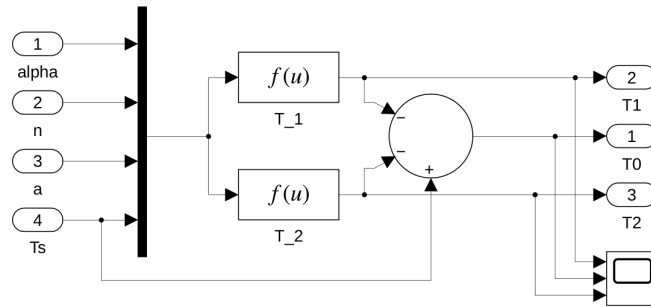


Figure 4.4: Dwell Time (T_1, T_2, T_0) Calculation Subsystem

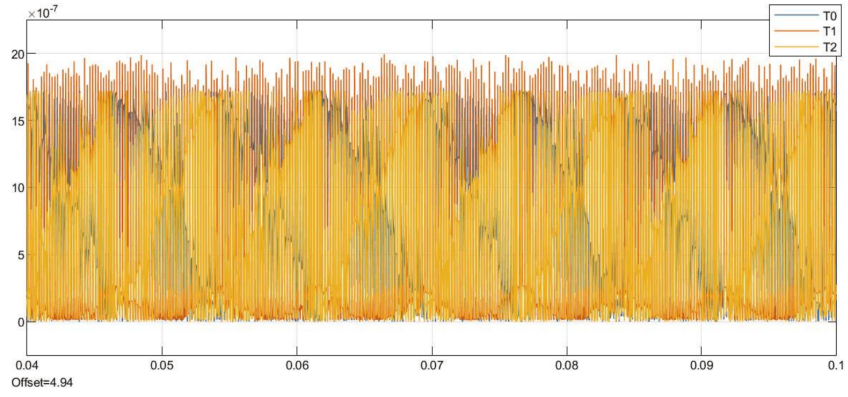


Figure 4.5: Calculated Dwell Times across Switching Periods

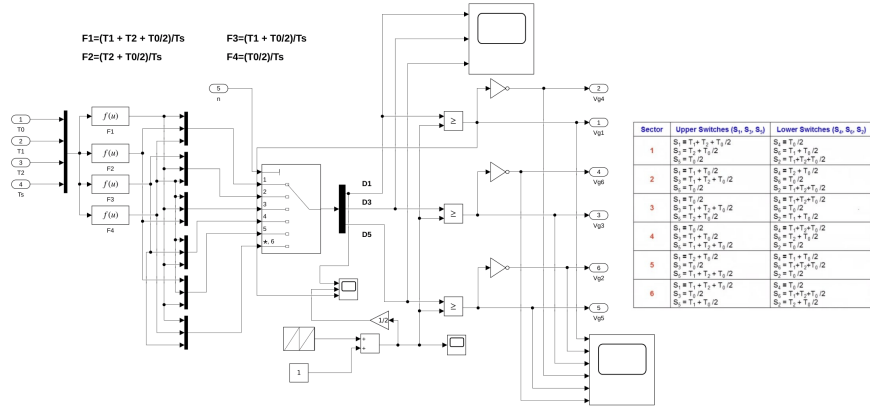


Figure 4.6: Detailed PWM Logic and Sector Lookup

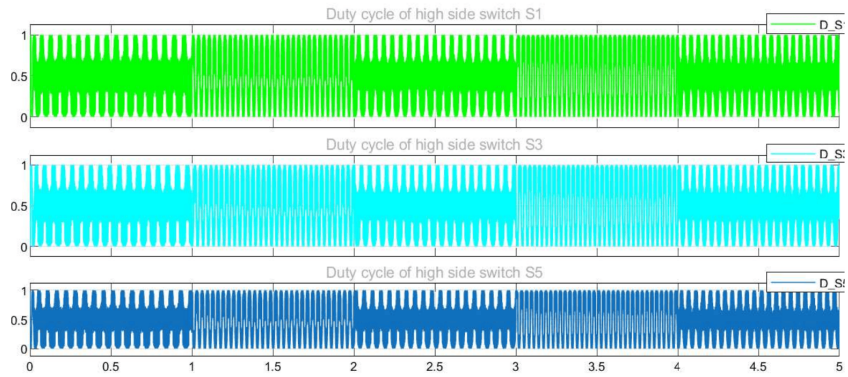


Figure 4.7: Duty Cycles of High-Side Inverter Switches

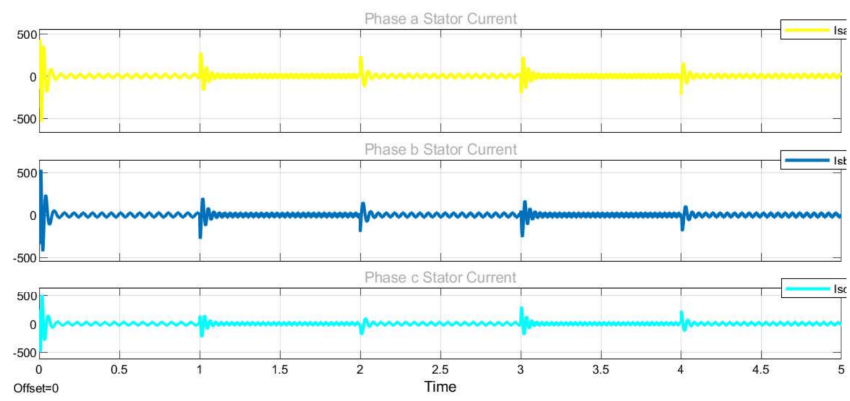


Figure 4.8: Three-Phase Stator Currents Responding to Load Transients

Chapter 5

Conclusions and Future Scope

5.1 Conclusions

The provided Simulink model serves as a functionally complete "digital twin" of a modern, industrial-grade Indirect Field Oriented Control (IFOC) system for a Squirrel-Cage Induction Motor. By meticulously modelling the electromechanical plant, coordinate transformations, and discrete-time control logic, this architecture demystifies complex, non-linear dynamics inherent in three-phase AC machines.

The architecture proves that an induction motor can be controlled with the same orthogonal precision and dynamic transient response as a separately excited DC motor. Furthermore, operating at a stringent $2\ \mu s$ sample time faithfully replicates the execution environment of a modern Digital Signal Processor (DSP).

5.2 Future Scope

Future enhancements to this digital architecture could include the implementation of sensorless speed observers (eliminating the need for physical encoders), integration of online parameter estimation to counteract thermal detuning of the rotor resistance, and advanced overmodulation strategies to further extend the field-weakening operating range of the induction motor.

Bibliography

- [1] Karmakar, J. (2026). *Comprehensive Analysis of Field Oriented Control (FOC) for a 3-Phase Induction Motor*. Cooch Behar Government Engineering College.